PART I

Projector Resolved Seed Calculus I

Typed operator systems, seed rank, and exact packet coordinates

Seeds, operators and projectors provide the linear language of the series. Exact packet coordinates make the generated objects and the maps between them visible.

IN THIS PART

Mathematical monograph 2

Machine-native paper 102

Lean and evidence guide 201

Scope of this part

The correspondence concerns the stated proof-level constructions. Noncomputable existence results do not provide executable solvers.

PART II

Projector Resolved Seed Calculus II

Refinement towers, weighted transfer, and integral separation of operator packets

Refinement towers compare packet descriptions across levels. Weighted transfers and integral separation show which structures survive a change of resolution.

IN THIS PART

Mathematical monograph 246

Visual guide 327

Lean and evidence guide 353

Scope of this part

Canonical values are defined for admissible data. A general existence theorem for admissible contours remains a separate question.

PART III

Projector Resolved Seed Calculus III

Typed transfer, nonlinear packets, and obstruction theory

Typed operations extend the linear language to nonlinear coordinates, analytic packet jets and obstruction theory. The analytic supplement develops the continuous packet-jet claims.

IN THIS PART

Mathematical monograph 405

Analytic packet jets and exact composition 504

Visual guide 510

Lean and evidence guide 536

Release record 588

Scope of this part

Expanded ordinary proofs and the bounded Lean core have different coverage. The packet Taylor estimates, counting arguments and later extensions retain their stated ordinary-proof status. Arbitrary differentiability is not asserted.

PART IV

Projector Resolved Seed Calculus IV

Cyclic analytic functions, reciprocal families, and spectral zero data

A cyclic analytic family connects coefficient selection, differential transport, reciprocal polynomials and spectral zero data.

IN THIS PART

Mathematical monograph 591

Visual guide 670

Lean and evidence guide 698

Verification closeout 831

Scope of this part

The analytic statements retain their branch choices, multiplicities and integer-order hypotheses. Universal zero simplicity and a zeta functional equation remain outside these results.

PART V

Multiresolution Signal Calculus

Exact coordinates, singular packets, and stable reconstruction

Signal models move through filters, packet trees and reconstruction maps. Exact coordinates and singular packets distinguish representation from approximation and recovery.

IN THIS PART

Mathematical monograph 837

Visual guide 972

Lean and evidence guide 1015

Scope of this part

Separated-jump results require the stated model, gap and grid assumptions. Stable edge-packet recovery and the broader approximation class remain open.

PART VI

Transfer Semigroups and Memory

Probabilistic refinement, fractional operators, and exact hidden-state elimination

Evolution, probability and conditioning share a transfer language. Eliminating hidden states produces memory terms whose hypotheses remain part of the model.

IN THIS PART

Mathematical monograph 1074

Visual guide 1229

Lean and evidence guide 1273

Scope of this part

Finite-state, common-gap and fixed-partition restrictions remain explicit. Completion, uniform refinement control and stable infinite latent-packet closure remain open. Floating residuals are not interval certificates.

PART VII

Quantum Seeds and Effective Dynamics

Symmetry, geometric transport, quantum channels, and bounded refinement towers

Quantum seeds generate spaces through an operator grammar. Symmetry, circuits, geometric transport and quantum channels lead to finite effective dynamics and bounded refinement questions.

IN THIS PART

Mathematical monograph 1338

Visual guide 1500

Lean and evidence guide 1542

Scope of this part

The finite formal contract does not certify the full continuum physics program. Anomaly-free refinement, uniform Feshbach control, Lorentzian reconstruction and the proposed gravitational limits remain open.

PART VIII

Packet Operator Algorithms

Exact compilation and GPU realization

Structured packet representations remove work before execution. Exact compilation connects algebraic structure, recompression, precision and stated hardware observations.

IN THIS PART

Mathematical monograph 1627

Visual guide 1753

Lean and evidence guide 1787

Scope of this part

Small-core optimality uses supplied ordered thin-SVD equations and actual compact ranges. General numerical SVD verification, unrestricted discovery and CUDA-kernel verification require further results.

PART IX

Euler Congruences

Completion, factorial strengthening, and sharp dyadic geometry

Euler congruences are developed through completion, factorial strengthening and dyadic geometry. The accompanying guides connect exact statements, bounded examples and formal coverage.

IN THIS PART

Mathematical monograph 1879

Visual guide 2119

Lean and evidence guide 2144

Scope of this part

The arithmetic, dyadic and lifting results have distinct hypotheses. General Euler-tail simplicity and the broader native and polyphase extensions remain open.

PART X

Universal Residual Seeds

Deformation forests, transport categories, and conditional system completion

Residual seeds, deformation forests and transport categories organize conditional system completion. Supplied arithmetic and carrier data remain explicit inputs.

IN THIS PART

Mathematical monograph 2277

Visual guide 2370

Lean and evidence guide 2403

Scope of this part

The target family and recovered carrier are independent existential outputs. Identifying them and proving nonzero occupancy require additional premises. Source families, carrier presentations and local comparisons remain supplied.

PART XI

Tame and Wild Packet Geometry

For local Langlands

Tame and wild packet geometry organizes the local arithmetic constructions. The research companions retain both restricted progress and the unresolved obligations of Conjecture 66.

IN THIS PART

Mathematical monograph 2441

Visual guide 2551

Lean and evidence guide 2604

Toward Arithmetic Packet Realization 2844

Arithmetic Packet Realization: visual atlas 2921

Scope of this part

Conjecture 66 remains open. Its unresolved requirements concern general arithmetic realization, H5 for the same candidate, corrected full-character occurrence and actual native data under the stronger contract.

PART XII

Patching Support and Hecke Packets

Conditional automorphic completion

Patching support and compatible Hecke packets complete a conditional arithmetic argument. Each application must supply the geometric, automorphic and representation-theoretic inputs that the theorem uses.

IN THIS PART

Mathematical monograph 2958

Visual guide 2998

Lean and evidence guide 3026

Scope of this part

Actual residual and deformation data, applicable patching and full support, a compatible same-field Hecke point, classicality and an attached representation are required. Comparisons of compatible families assume those families already exist. Paper XI Conjecture 66 is not assumed proved.